

Leaflet + MongoDB + Nominatim Notes

1. Setup Process

1. **Install dependencies** (Express project example):

```
npm install express mongoose axios ejs connect-flash express-session multer
```

2. **Include Leaflet in your EJS template** via CDN:

```
<!-- Leaflet CSS -->
<link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/
leaflet.css" />
<!-- Leaflet JS -->
<script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
```

3. **Set up MongoDB schema** for listing with geometry:

```
const listingSchema = new mongoose.Schema({
  title: String,
  description: String,
  location: String,
  country: String,
  geometry: {
    type: { type: String, enum: ['Point'], required: true },
    coordinates: { type: [Number], required: true } // [lng, lat]
  },
  image: { url: String, filename: String },
  owner: { type: mongoose.Schema.Types.ObjectId, ref: 'User' }
});
```

4. **Fetch coordinates using Nominatim** and store them:

```
const axios = require('axios');
const geoRes = await axios.get('https://nominatim.openstreetmap.org/
search', {
  params: { q: `${location}, ${country}`, format: 'json', limit: 1 }
});
```

```
const coordinates = [parseFloat(geoRes.data[0].lon),
parseFloat(geoRes.data[0].lat)];
```

5. **Pass coordinates to EJS template** when rendering:

```
res.render('listing.ejs', { list, coordinates });
```

6. **Render map in EJS** using Leaflet:

```
<div id="map" style="height: 400px;"></div>
<script>
  const coords = <%= JSON.stringify(coordinates) %>; // [lng, lat]
  const map = L.map('map').setView([coords[1], coords[0]], 13); // swap for
  Leaflet

  L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
    attribution: '&copy; <a href="https://www.openstreetmap.org/"
copyright">OpenStreetMap</a>'
  }).addTo(map);

  L.marker([coords[1], coords[0]]) // swap again
    .addTo(map)
    .bindPopup(`<b><%= list.title %></b><br><%= list.location %>`)
    .openPopup();
</script>
```

2. Coordinate Formats

- **GeoJSON (MongoDB storage):** [longitude, latitude] (lng first)
- **Leaflet (map display):** [latitude, longitude] (lat first)

⚠ Swap coordinates when passing to Leaflet.

3. Fetching Coordinates from Nominatim

```
const geoRes = await axios.get('https://nominatim.openstreetmap.org/search', {
  params: { q: `${location}, ${country}`, format: 'json', limit: 1 }
});
```

- Always validate response before storing. - Use `parseFloat` to convert strings to numbers.

4. Storing in MongoDB

```
const newListing = new Listing({
  title, description, price, location, country,
  geometry: { type: 'Point', coordinates }, owner: req.user._id
});
```

- Coordinates stored as `[lng, lat]`.

5. Displaying Map with Leaflet

- Only render map if coordinates exist.
 - Swap `[lng, lat]` to `[lat, lng]` for Leaflet display.
 - Include Leaflet CSS & JS via CDN.
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6. Common Mistakes

- Using `import L from 'leaflet'` in EJS ❌.
 - Not swapping coordinates → wrong marker location.
 - Storing coordinates as strings.
 - Using default location → loses actual data.
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7. Best Practices

- Validate Nominatim response.
- Store coordinates as numbers `[lng, lat]`.
- Swap coordinates only for Leaflet display.
- Show message if coordinates missing.
- Include Leaflet CSS & JS before using `L`.